

Oliver Krisetya

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EDUCATION

The George Washington University
Combined B.S./M.S. in Computer Science
Music Minor

Washington, DC
May 2026

- Technical GPA: 3.76 - Dean's List - Recipient of Stephen Joel Trachtenberg Full-Ride Scholarship (2021)
- Design Build Fly, Rocket Team

Relevant Courses: Advanced Machine Learning, Natural Language Understanding, Design of Interactive Multimedia, Artificial Intelligence, Advanced Operating Systems, Systems Programming, Computer Architecture, Data Structures & Algorithms

SKILLS

Languages: Python, C, Java, JavaScript, TypeScript, SQL, Bash
Systems: Linux/Unix, POSIX APIs, Networking, Concurrency, Debugging

Tools: Git, Docker, Make, GDB, Valgrind, Github, Github Classroom
Frameworks/Databases: PyTorch, React, PostgreSQL, MySQL, SQLite

PROJECTS

George Washington University - Department of Computer Science
Minilab Research Project

Washington, DC
May 2025 - December 2025

- Conducting **comparative evaluation and data validation** of binary size-reduction techniques for **embedded systems**, analyzing the effectiveness of competing ELF-shrinking methods
- Designed and executed a **benchmarking framework** to measure binary size changes across multiple tools and configurations, enabling apples-to-apples comparison

Ticket to Ride AI - Monte Carlo Tree Search Agent

- Designed an AI agent for a partially observable, stochastic multiplayer board game by formalizing the state space, actions, and transition model, then implementing a Monte Carlo Tree Search to select moves under uncertainty
- Implemented rollout-based simulation (1,000 simulations per turn) to estimate expected final score for each candidate move, avoiding exhausting search over an intractably large branching factor
- Built and compared two distinct reward functions (ticket-completion vs. incremental progress) to evaluate agent strategy trade-offs; supports both human-vs-AI and AI-vs-AI play

Register Bias in LLM Outputs - Measurement & Mitigation

- Built a dual-measurement framework pairing a custom lexical formality-rate metric with a fine-tuned RoBERTa formality classifier to quantify register bias in LLM outputs against human conversational baselines
- Evaluated formality drift across six model families (GPT, PaLM, LLaMA, Mistral, Claude, Vicuna) using the LMSYS-Chat-1m dataset and human corpora (Reddit, BlendedSkillTalk, ELI5)
- Designed and trained a LoRA fine-tuning pipeline on Qwen2.5-3B-Instruct that reduced formal register markers by 12% while preserving base fluency (stable ROUGE-L)

Easy-Sched - Automated LA/UTA Scheduling Platform

- Designed and implemented a constraint satisfaction solver (backtracking with MRV and branch-and-bound pruning) to auto-assign Learning Assistants and UTAs to course sections while maximizing overall fit
- Separated hard eligibility constraints (GPA floors, scheduling conflicts) from soft fit scoring as independent systems, allowing scoring schemes to be swapped without touching the solver
- Built a typed React/TypeScript frontend with a drag-and-drop assignment board and isolate per-workspace sandboxes for side-by-side scenario comparison

EXPERIENCE

George Washington University - Department of Computer Science
Graduate Teaching Assistant

Washington, DC
August 2025 - May 2026

- Authored C projects and assessments for a graduate Computer Systems Fundamentals course, and built/maintained autograders, text harnesses, and GitHub Classroom workflows for reproducible grading
- Led curriculum development for an undergraduate Databases & Team Projects course, including lab design and coordination of two undergraduate TAs

Undergraduate Teaching Assistant

August 2024 - May 2025

- Supported **Operating Systems** and **Computer Architecture** courses, debugging C and Assembly programs, and clarifying low-level execution and memory concepts.
- Designed and validated homework test cases and lab materials, and led lab sessions and office hours

Learning Assistant

August 2022 - May 2024

George Washington University Esports
President

Washington, DC
January 2025 - May 2026

- Led a **200-member organization across 12 competitive and casual teams**, oversaw budgeting, scheduling, recruitment, and strategic planning.
- Managed and coordinated **five executive officers and additional board members**, establishing operational structure and accountability across departments.

Executive of Competition (Jan 2023 - Jan 2025) | **Competitive Organizer** (Aug 2021 - Jan 2023)